

Motilal Nehru National Institute of Technology Allahabad
Department of Computer Science and Engineering

Introduction to Artificial Intelligence and Machine Learning (CSN12601)

Lab Manual: Assignment-6

Lab Title: Programs on File Handling

Objective:

1. To understand the basics of file handling in Python, including reading, writing, and manipulating files.
2. To practice reading data from various file formats such as text, CSV, Excel, TSV, and image files, highlighting Python's versatility in handling different types of data.
3. To learn how to process and manipulate file contents, including operations like sorting words alphabetically and copying content from one file to another.

Materials Required:

- Computer with internet access
- Python (latest version)
- Python Integrated Development Environment (IDLE)
- Access to an online Python environment (e.g., Google Colaboratory)

Theory:

File Handling in Python: File handling is a crucial aspect of programming that involves performing operations on files such as reading, writing, and appending data. Python provides built-in functions and modules like `open()`, `read()`, `write()`, and `close()` to perform these operations with ease. Understanding file handling is essential for tasks that involve data storage, retrieval, and manipulation.

1. **Reading and Writing Files:**

    - **Read Operation (`read`, `readline`, `readlines`):** These methods allow you to read content from a file. `read()` reads the entire file as a single string, `readline()` reads one line at a time, and `readlines()` returns a list of all lines in the file.
    - **Write Operation (`write`, `writelines`):** These methods allow writing data to a file. Using `write()`, you can write a string to a file, while `writelines()` writes a list of strings.

2. **Unique Words in Alphabetical Order:**

    - Reading a text file and identifying unique words involves reading the content, splitting the text into words, removing duplicates using a set, and sorting the words alphabetically. This helps in understanding text processing and manipulation.

3. **Copying File Contents:**

    - Copying contents from one file to another involves reading from the source file and writing to the destination file. This process demonstrates the practical use of file handling functions and emphasizes the importance of file operations in data management.

4. **Handling Different File Formats:**

    - **CSV Files:** Comma-Separated Values (CSV) files are read using Python's `csv` module, which allows you to easily parse and write CSV data.
    - **Excel Files:** Reading Excel files requires modules like `pandas` or `openpyxl`, which can read and manipulate Excel worksheets efficiently.
    - **Text Files:** Text files can be managed using basic file handling functions like `open()`, with modes like `r` for reading, `w` for writing, and `a` for appending.
    - **TSV Files:** Tab-Separated Values (TSV) files are similar to CSV but use tabs as delimiters. They can also be read using the `csv` module with appropriate delimiter settings.
    - **Image Files (JPG):** Handling image files like JPG requires libraries such as `PIL` (Python Imaging Library) or `OpenCV`, which allow reading, displaying, and processing image data.

List of Programs in Assignment-6

1) Write a program that inputs a text file. The program should print all of the unique words in the file in alphabetical order.
2) Write a Python program to perform read and write operations on a file.
3) Write a Python program to copy the contents of one file to another file.
4) Write a Program to read csv, excel, txt, tsv, and jpg file in Python.